Effects of Container Type and Silica Gel Amount on Cat Food Storage

30 g of cat food per container; 3 independent, repeated containers per treatment

Abstract

This experiment is based on the environment in our automatic cat food machine project. It uses loss-on-drying (LoD), pH, and an iodine spot test to assess cat food storage under different silica gel amounts in plastic and foil containers. Silica gel lowered the LoD in both containers on D14, but on D21 the effect only remained in the foil groups, where LoD decreased with more silica gel (0 g, 7.60%;1 g, 6.70%; 2 g, 6.67%; 4 g, 4.50%). Without silica gel, foil did not have a lower LoD than plastic (7.60% vs 7.13% on D21). The pH was slightly lower in the foil groups than in the plastic groups across different amounts of silica gel on D14 (0.05-0.09), and on D21 it was significantly lower only at 0 g and 4 g. In the spot test on D21, the foil 1 g and 2 g groups had significantly higher integrated intensity than the foil control (0 g,3964;1 g,5003;2 g,5787), while the plastic groups showed no difference. However, the foil control had the lowest signal of all groups, and the foil silica gel groups were similar to the plastic control. The results suggest a possible link between moisture and lipid-soluble components, but adding more silica gel (4 g) did not significantly increase the spot signal, even though it lowered LoD the most. The pH results suggest that the pH differences were likely caused by the container material rather than by silica gel. The hypothesis that more silica gel would better lower moisture and preserve lipid-soluble components in a plastic container was not supported. For the automatic feeder, silica gel in a plastic box may need to be replaced about every two weeks, while a foil-lined box with silica gel may keep the food drier for three weeks.

Introduction

A major issue occurs when cat food is exposed to air for a long time: the texture of the cat food becomes soft and may turn sour, which cats may dislike. During long-term storage, the cat food absorbs moisture from the surrounding environment, and its antioxidant ability decreases over time. Lipids are one of the major nutrients and additives in cat food, and they include lipid-soluble antioxidants such as vitamin E. Under long-term exposure to air, oxidation and an increase in moisture lead to an overall decrease in the quality of cat food (Silva et al., 2025).

High moisture makes it easier for mold and bacteria to grow on the cat food; triglycerides in lipids are broken down by hydrolysis into fatty acids, which drive the pH down on their own. Lipids also react with oxygen through oxidation reactions, a process that is accelerated by factors such as light, heat, and degree of exposure to air. Oxidized lipids can cause the food to have a rotten smell and taste. One molecule that plays a significant role in preventing oxidation and is an essential nutrient for cats is vitamin E, which donates hydrogen to free radicals and stops the lipid oxidation chain reaction (National research Council 2006; Ralla et al.,2024). The protection process also consumes vitamin E over time. However, in this experiment, vitamin E cannot be measured directly because iodine also stains many other lipid-soluble compounds. The spot signal can compare lipid-soluble extracts under different storage conditions, but it cannot measure the amount of vitamin E or other nutrients preserved.

Automatic feeders are commonly used by people to save their time and energy, especially for people who are not at home or are on trips. In these cases, people only need to pour cat food into the feeder and set it to feed the cat automatically. However, automatic distribution requires cat food to stay in the storage box. The airflow during refills and distribution, and the gaps between the lid and the walls, expose a large amount of cat food to air, which creates a new storage issue.

Plastic is the most commonly used material for cat feeder storage boxes, due to its low price and convenience. Aluminum foil was also tested because it is commonly seen in kitchens and blocks light exposure. Furthermore, silica gel is a desiccant; it can absorb water and keep the surrounding environment dry for a certain period of time.

This study investigates the effects of plastic and foil containers with different amounts of silica gel on the storage conditions of cat food, and how this applies to the automatic feeder we designed. This study asks three questions: whether adding silica gel has an effect; whether more silica gel has better preservation of the cat food; and whether different containers affect the preservation. I hypothesized that more silica gel would better lower moisture and help preserve lipid-soluble components in a plastic container.

Methods

1. Groups and Storage

Royal cat food was used for the entire experiment in two types of containers: plastic and aluminum foil. Each container was treated with 0, 1, 2, or 4 g of silica gel, with 0 as the control. Each treatment had 3 replicate containers with 30 g of cat food each, for a total of 8 treatments and 24 containers. The cat

food was stored in a basement (~65% humidity, ~22°C). The relative humidity was 74% on Day 0, 64% on Day 14, and 65% on Day 21.

2. Loss on Drying (LoD)

The empty aluminum dish was weighed first (M0). About 1g of cat food was added, and the loaded weight of the cat food and dish was weighed before heating (M1). The sample was heated to 250°F for 3 h, cooled for 20 min, then weighed as M2. Each container was tested once on each test day. LoD = (M1-M2)/(M1-M0) x 100%.

The higher LoD indicates the larger fraction of mass was lost during the heating. This is used as a water indicator. However, the heating may also remove other volatile substances; so it cannot exactly and accurate to prove the moisture concentration.

3. Iodine Spot Test: Lipid-Soluble Signal

0.5 g of cat food was ground and mixed with 2.5 mL of hexane, and the tube was capped for 20 min away from light. The extract was separated, and another 2.5 mL of hexane was added, then removed in the same way. The plate was dipped in hexane: ethyl acetate 9:1 and dried. After the cat food extract was spotted on the plate without development, it was stained with iodine solution and photographed with a phone; ImageJ was used to calculate the integrated intensity of the selected spot area, which reflects the color intensity, but this can't be used as a tool to determine the concentration of Vitamin E or total fat remaining.

Each group had 6 lanes on the plate. Adjacent lanes came from the same container for replicate groups. Therefore, during the data analysis, the data were averaged and tested with 3 independent, same-condition groups. Lane 6 of the D14 plastic 1g group was aberrant; it was excluded from the data.

4. pH

1 g of cat food for each container was mixed with 10 ml of distilled water and left for 20 min. The pH of the water extract was then measured with a pH meter calibrated with a pH 7 buffer. Each sample was read twice, and the two readings were averaged; the electrode was rinsed and wiped after each test. pH was measured on D0, D14, and D21; the plastic group was measured first, then the foil groups. Since the calibration record is not stable, the data can only be compared within the same day.

5. Statistics

Each treatment had 3 independent storage containers, so the analysis used n = 3. In the spot test, the adjacent two lanes were averaged. For LoD, each container was measured once. Silica gel groups were compared with the 0 g group of the same type of container on the same day, and the foil was compared with plastic at the same dose. Welch's t-test was used. $P<0.05$ was considered significant; not significant means there was not enough information to support differences.

Spot test plates were run on different days. It is suggested in this experiment to compare only with the results on the same day. The relationship between LoD and spot signal on D21 was tested with Pearson correlation using the 8 group means.

Results

Loss on Drying (LoD)

Table 1. Mean LoD of each group and comparison with control

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Treatment	D0 LoD (%)	D14 LoD (%)	D14 p	D21 LoD (%)	D21 p
Plastic 0 g	4.50	7.18	—	7.13	—
Plastic 1 g	5.90	3.05	<0.001	7.27	0.730
Plastic 2 g	5.97	6.20	0.016	7.27	0.735
Plastic 4 g	5.50	5.35	0.004	7.10	0.939
Foil 0 g	4.80	8.06	—	7.60	—
Foil 1 g	4.20	5.73	0.003	6.70	0.027
Foil 2 g	5.33	1.76	<0.001	6.67	0.023
Foil 4 g	4.80	2.30	<0.001	4.50	<0.001
n = 3 containers. p: Welch t-test					

- LoD was calculated from the weighing records and rounded to two decimals; P is compared with 0g groups.
- On D14, all silica gel groups had significantly lower LoD than their own controls.
- On D21, the plastic silica gel groups were no longer significantly different from the control, and all foil silica gel groups were still lower ($P<0.05$), and LoD decreased as silica gel increased (7.60%, 6.70%, 6.67%,4.50%).
- The control LoD increased from about 4.5-4.8% on D0 to 7%-8% on D14 and D21, showing that the cat food absorbed moisture during storage.
- All six silica gel groups had lower LoD on D 14 but rose by D 21(e.g., foil 2 g: 1.76% to 6.67%), suggesting the silica gel became saturated.

- On D0, all groups had the same food, so the D0 range (4.2-5.97%) shows the measurement variation. Without silica gel, foil did not have a lower LoD than plastic on D14(8.06% vs 7.18%) or D21(7.60% vs 7.13%).

Spot test lipid- soluble signal

Table2

Table 2. Spot test signal by container and test results					
Treatment	D14 mean signal ± S	D14 p		D21 mean signal ± S	D21 p
Foil 0 g	11,496 ± 2,320	—		3,964 ± 96	—
Foil 1 g	14,190 ± 758		0.173	5,003 ± 414	0.043
Foil 2 g	14,316 ± 929		0.159	5,787 ± 344	0.008
Foil 4 g	12,784 ± 2,136		0.518	5,345 ± 1,001	0.138
Plastic 0 g	12,049 ± 914	—		5,012 ± 670	—
Plastic 1 g	13,623 ± 338		0.082	4,380 ± 550	0.278
Plastic 2 g	13,829 ± 373		0.062	4,787 ± 632	0.695
Plastic 4 g	16,257 ± 511		0.005	4,564 ± 634	0.448
n = 3 containers (adjacent lanes averaged). p: Welch t-test					

- On D14, all foil silica gel groups had higher means, but none were significant, partly because the foil control varied a lot between containers (± 2,320). Plastic groups: only 4 g was significantly higher than the control.
- On D21, foil 1 g and 2 g were significantly higher than the control, while 4 g was not; no plastic group differed from the control. The signal did not increase further from 2 g to 4 g.
- The foil control (3964) had the lowest signal of all 8 groups, and the foil silica gel groups were not significantly different from the plastic control(P = 0.985, 0.173, 0.660 for 1, 2, 4 g).
- Across the 8 groups on D21, a higher LoD was associated with a lower spot signal (r= -0.59), but this was not significant (P = 0.126). D14 and D21 were measured on different plates and cannot be compared.

Spot Signal Container Comparison

Table3

Table 3. D21 foil groups compared with plastic groups at the same dose			
Silica gel	Foil vs. plastic mean	p	
0 g	-21%	0.111	
1 g	+14%	0.197	
2 g	+21%	0.093	
4 g	+17%	0.328	
Positive - foil mean was higher. p: Welch t-test, n = 3 containers.			

- On D21, foil and plastic were not significantly different at any dose. On D14, they were also not different (P=0.73 ,0.33 ,0.47 ,0.10 for 0 - 4g).

pH

Table 4.

Table 4. Mean pH of each group and foil vs. plastic comparison at the same dose					
Treatment	D0 pH	D14 pH	D14 p	D21 pH	D21 p
Plastic 0 g	5.85	5.80	—	5.84	—
Plastic 1 g	5.86	5.79	—	5.83	—
Plastic 2 g	5.87	5.76	—	5.79	—
Plastic 4 g	5.86	5.77	—	5.79	—
Foil 0 g	5.87	5.71	<0.001	5.79	0.009
Foil 1 g	5.87	5.71	0.006	5.76	0.197
Foil 2 g	5.87	5.71	0.013	5.80	0.877
Foil 4 g	5.85	5.72	0.093	5.76	0.024
n = 3 containers (two readings averaged per container). p: Welch t-test, foil : plastic					

- D0, there was no pH difference between the two containers (5.85-5.87). On D14, the foil groups had about 0.065 lower pH than the plastic. It is significant at 0,1, 2 g, but not significant for 4 g.
- On D21, the foil groups were lower, but the differences are smaller than on D14 (0.035); 1 g and 2g have no significant differences.
- Within the same container, silica gel did not clearly change pH.

- The drop and rise in pH may be caused by errors in calibration records; the comparison and results across days may not be reliable, but the data from the same day can be used to refer to the effect of different containers and silica gel.

Conclusions

Silica gel can lower the moisture in the first two weeks, but by D21 the difference from the control was not significant in the plastic container. In the foil container, silica gel still lowered LoD on D21, and more silica gel gave a lower LoD. Its effect may depend on the ratio between the silica gel and the cat food, as well as the house humidity, because it can become saturated quickly if the humidity is high. The LoD of all silica gel groups rose from D14 to D21, which supports this. Foil alone did not keep the cat food drier.

In the foil container, the Spot signals with 1,2,4g of silica gel were 26%, 46%, and 35% higher than the control on D21. But only 1g and 2 g were significant. This may also be due to the lower LoD later on; longer-lasting moisture control may help preserve some lipid-soluble components. However, the foil silica gel groups were not higher than the plastic control. plastic container with silica didn't have the advantage of preserving the spot signal. The increase seen in plastic 4 g on D14 was not significant on D21. The LoD and spot patterns suggest a possible link between moisture control and the lipid-soluble compounds preservation. But the correlation was not significant ($r=-0.59$, $P=0.126$).

The foil groups had a slightly lower pH than the plastic groups at most doses. There was no difference on D0, but on D14 and D21, the same-dose comparison was significant at 3 of 4 doses on D14 and 2 of 4 doses on D21. The foil silica groups had a lower LoD but similar pH regardless of silica, which suggests pH did not simply follow LoD and was likely caused by the container.

The results do not support my hypothesis that more silica gel would better decrease moisture and preserve lipid-soluble compounds in a plastic container.

These results predict that a plastic storage container with silica gel will lose its moisture control within about two weeks in a humid basement, so the silica gel needs to be replaced about every two weeks. A foil container with about 2 g of silica gel per 30g of cat food may keep the food drier for three weeks. In addition. Adding more than 2 g of silica gel may make the food drier but may not support better preservation of lipid-soluble components. Future studies can test the saturation at different humidities and combine it with other methods to help preserve the food for a longer period.

Reference

National Research Council. (2006). *Nutrient requirements of dogs and cats*. The National Academies Press. https://doi.org/10.17226/10668

Ralla, T., et al. (2024). Over 100 years of vitamin E: An overview from synthesis and formulation to application in animal nutrition. *Journal of Animal Physiology and Animal Nutrition*. https://doi.org/10.1111/jpn.13919

Silva, A. M., Zangirolami, M., Assakawa, A. C., Santos, P. D. S., Alves, E. S., Vasconcellos, R. S., & Santos, O. O. (2025). Lipid quality and oxidative stability in industrial foods for dogs and cats. *Journal of the Brazilian Chemical Society*, *36*(12), Article e-20250161. https://doi.org/10.21577/0103-5053.20250161